

Lab: Breaking Functional Fixedness and Perceiving Latent Needs

Objective

Practice the perceptual skills that underpin creative invention: detecting prediction errors, overcoming functional fixedness, recognizing cross-domain patterns, and perceiving latent needs. By the end of this lab, you will have applied multiple perceptual reframing techniques to your own invention idea and identified at least one opportunity you had not previously seen.

Materials / Prerequisites

- Your own invention idea or project (any domain, any stage)
- One common household object (a pen, a mug, a fork, a rubber band — anything within reach)
- Paper and pen for sketching
- A timer
- Completed reading of Module 03: Creative Perception

Part 1: The Generic Parts Technique (10 minutes)

Functional fixedness locks you into seeing objects by their conventional function. The Generic Parts Technique breaks this by describing objects purely by physical properties. This technique was developed by cognitive scientists studying creative problem solving and has been shown to significantly increase the number and novelty of alternative uses people can generate for common objects. The key mechanism is simple: when you remove the functional label from an object, you reduce the precision of your categorical priors about what the object "is for," making alternative functions perceptible. This is not a metaphorical exercise — it produces measurable changes in creative output.

Step 1 (3 min): Take the household object you selected. Write its conventional name and function:

Object name:

Write your response here

Conventional function:

Write your response here

Step 2 (3 min): Now describe the SAME object using only physical properties — material, shape, dimensions, weight, texture, flexibility, conductivity, color. Do NOT mention its conventional function.

Physical-properties-only description:

Write your response here

Step 3 (4 min): Based solely on the physical description you just wrote, generate at least five alternative uses for this object. These should be plausible uses based on physical properties, not jokes.

Alternative Use	Which Physical Property Enables It

Write your response here

|

Write your response here

Write your response here

|

Write your response here

Write your response here

|

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

Reflection: How did stripping the conventional label change what you could see? What happened to your perception when you described the object without naming it?

Write your response here

Part 2: Cross-Domain Pattern Recognition (15 minutes)

Cross-domain pattern recognition is the ability to see structural similarities between systems in completely different domains. This capacity underlies many of history's most important inventions: the Wright brothers applied bird flight principles to airplane control, de Mestral applied burr hook-and-loop mechanisms to fastener design, and Nakatsu applied kingfisher beak morphology to bullet train nose design. The skill is not innate — it is developed by exposing yourself to diverse domains and practicing the abstraction of structural principles from specific instances. In this exercise, you will deliberately search for structural analogies to your invention problem across three different domains.

Step 1 (5 min): Describe your invention idea in one paragraph. Focus on the core PROBLEM you are trying to solve, not your current solution:

Write your response here

Step 2 (5 min): Now find structural analogies to your problem in three different domains. For each domain, identify a system that faces a structurally similar challenge and describe how it solves it:

Domain	Analogous System	The Similar Challenge	How It Solves It
Biology / Nature			

Write your response here

|

Write your response here

|

Write your response here

|| A completely different industry |

Write your response here

|

Write your response here

|

Write your response here

| | Everyday life / household |

Write your response here

|

Write your response here

|

Write your response here

|

Step 3 (5 min): Choose the most promising analogy from Step 2. Describe how the analogous solution could be translated to your invention domain. What would your invention look like if it solved the problem the way the analogous system does?

Selected analogy:

Write your response here

How it translates to my invention:

Write your response here

What new features or approaches does this translation suggest?

Write your response here

Part 3: Perceptual Reframing Exercise (15 minutes)

Perceptual reframing techniques work by deliberately modifying your generative model so that different prediction errors become visible. Each technique changes a different aspect of the model — inversion changes what outcomes are predicted, scale shifting changes which variables are salient, and stakeholder rotation changes whose preferences are being modeled. The goal is not to find one "right" reframe but to experience how the same problem looks completely different when perceived through a different model. Apply all three techniques below to your invention idea.

Reframe 1 — Inversion (5 min): Instead of asking "How can I solve this problem?", ask "How can I make this problem as bad as possible?" List at least five ways to make the problem worse:

- 1.

Write your response here

1.

Write your response here

1.

Write your response here

1.

Write your response here

1.

Write your response here

Now invert each "worse" idea. Does any inversion suggest a new solution approach?

Most promising inverted insight:

Write your response here

Reframe 2 — Scale Shift (5 min): Describe your problem at three different scales:

At the individual/component level:

Write your response here

At the system/organization level:

Write your response here

At the societal/civilizational level:

Write your response here

Which scale reveals the most interesting new features of the problem?

Write your response here

Reframe 3 — Stakeholder Rotation (5 min): Describe the problem from three different stakeholders' perspectives:

Stakeholder	How They Experience the Problem	What They Would Want From a Solution
Primary user		

Write your response here

|

Write your response here

| | Someone who benefits indirectly |

Write your response here

|

Write your response here

| | Someone who might resist the solution |

Write your response here

|

Write your response here

|

Did the stakeholder rotation reveal needs you had not considered?

Write your response here

Part 4: Latent Need Detection (12 minutes)

Latent needs are the most valuable type of need to address because they represent uncontested market space — no competitor is addressing them because no one has articulated them. Users themselves cannot express latent needs in surveys because their generative models have adapted to the status quo. The only way to detect latent needs is through careful observation of

actual behavior, paying attention to workarounds, frustrations, and the small adaptations people make unconsciously. Jane Fulton Suri of IDEO called these adaptations "thoughtless acts" — they are the visible traces of needs that have gone unmet for so long that people have stopped noticing them.

Step 1 (6 min): Think about the users or beneficiaries of your invention. Describe their current behavior — not what they say they want, but what they actually do. Focus on workarounds, frustrations, and "thoughtless acts" (unconscious adaptations to the current situation):

Observed Behavior / Workaround	What It Reveals About Their Actual Needs	Is This Need Currently Addressed?

Write your response here

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Write your response here

|

Write your response here

Write your response here

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Write your response here

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Write your response here

Write your response here

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Write your response here

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Write your response here

Write your response here

Write your response here

Write your response here

Step 2 (6 min): Identify the single most significant latent need you discovered — a need that users have but cannot articulate because they have adapted to the status quo:

The latent need:

Write your response here

Why users cannot articulate it:

Write your response here

How my invention could address it:

Write your response here

How does addressing this latent need change my invention concept?

Write your response here

Part 5: Integration (8 minutes)

This is the most important part of the lab. The perceptual techniques in Parts 1-4 generated raw material — new observations, new framings, new needs. Integration is the process of synthesizing this material into actionable insights that will change how you approach your invention going forward. Not everything you generated will be valuable, but the best labs typically produce two or three genuinely transformative insights — perceptions that permanently change the inventor's understanding of their problem. Review everything you

have generated in Parts 1-4. Identify the three most valuable perceptual insights — things you can now see about your invention idea that you could not see before this lab:

Insight 1:

Write your response here

Insight 2:

Write your response here

Insight 3:

Write your response here

Which perceptual reframing technique was most productive for you personally? Why do you think it worked better than the others for your specific problem?

Write your response here

Discussion / Debrief Questions

1. The Generic Parts Technique asks you to strip away conventional labels. How does this relate to the Active Inference concept of reducing precision on categorical priors? Did you feel the difference in how you perceived the object?
2. Cross-domain pattern recognition requires carrying generative models across domain boundaries. Which of your analogies felt most natural, and which felt most forced? What does this tell you about the breadth of your current generative models?
3. The three reframing techniques (inversion, scale shift, stakeholder rotation) each modify different aspects of the generative model. Which aspect of your model was most resistant to change? Why?
4. Latent need detection requires maintaining a model of what users would prefer if constraints were removed, rather than what they currently accept. How do you distinguish between a genuine latent need and a need you are projecting onto users?
5. How might you build these perceptual practices into your regular inventive routine? What would it look like to systematically practice creative perception, rather than waiting for insight to strike?